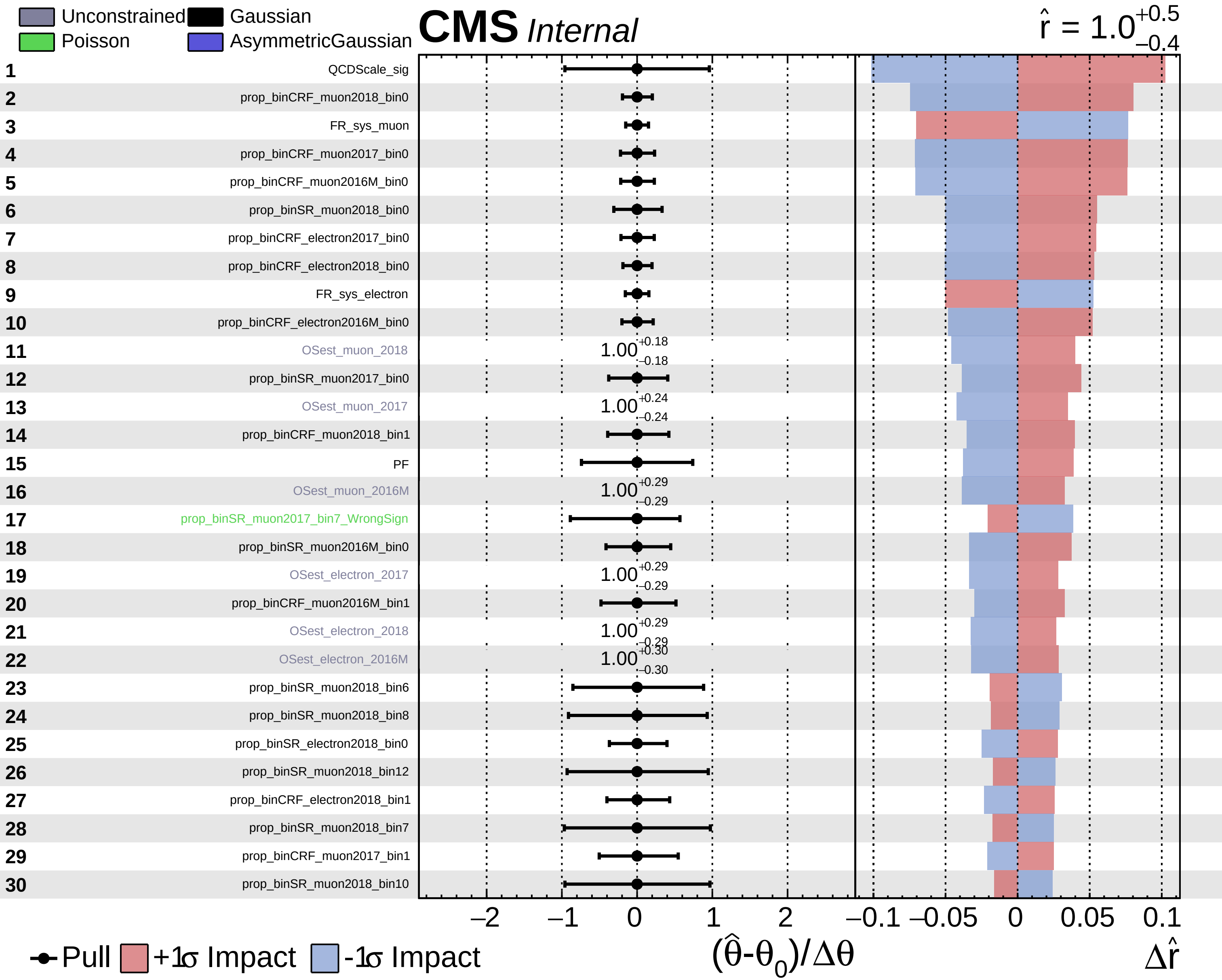
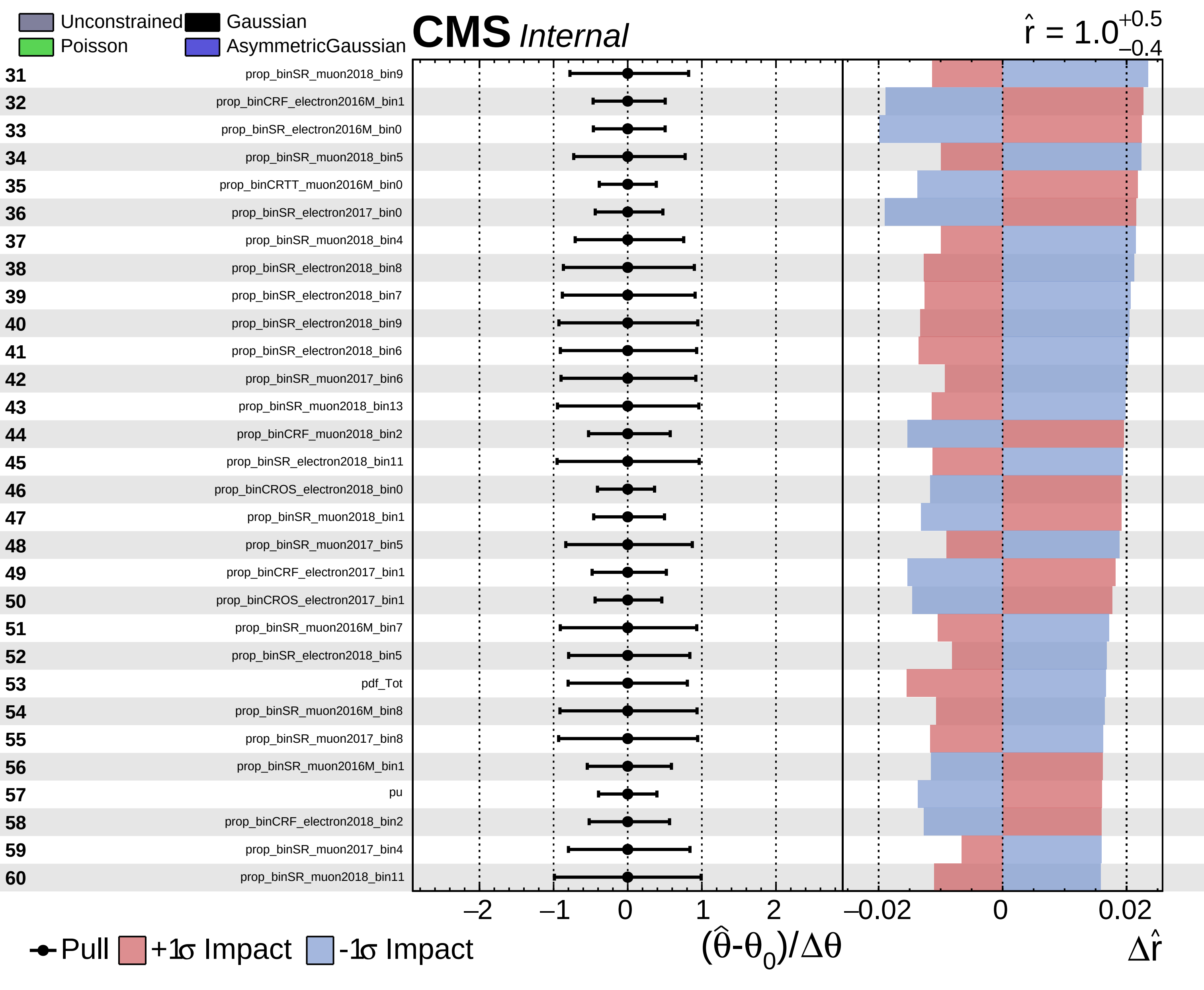
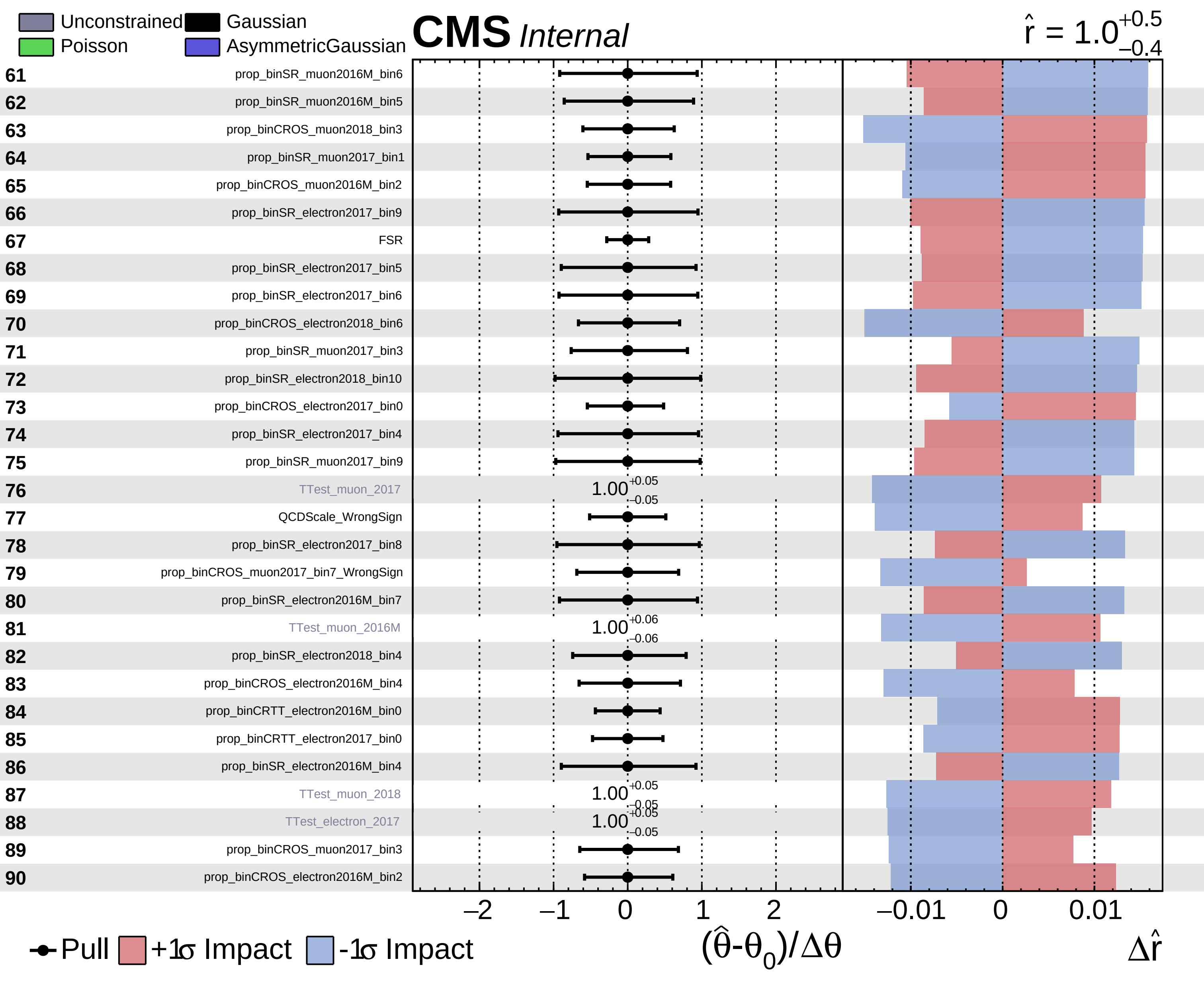


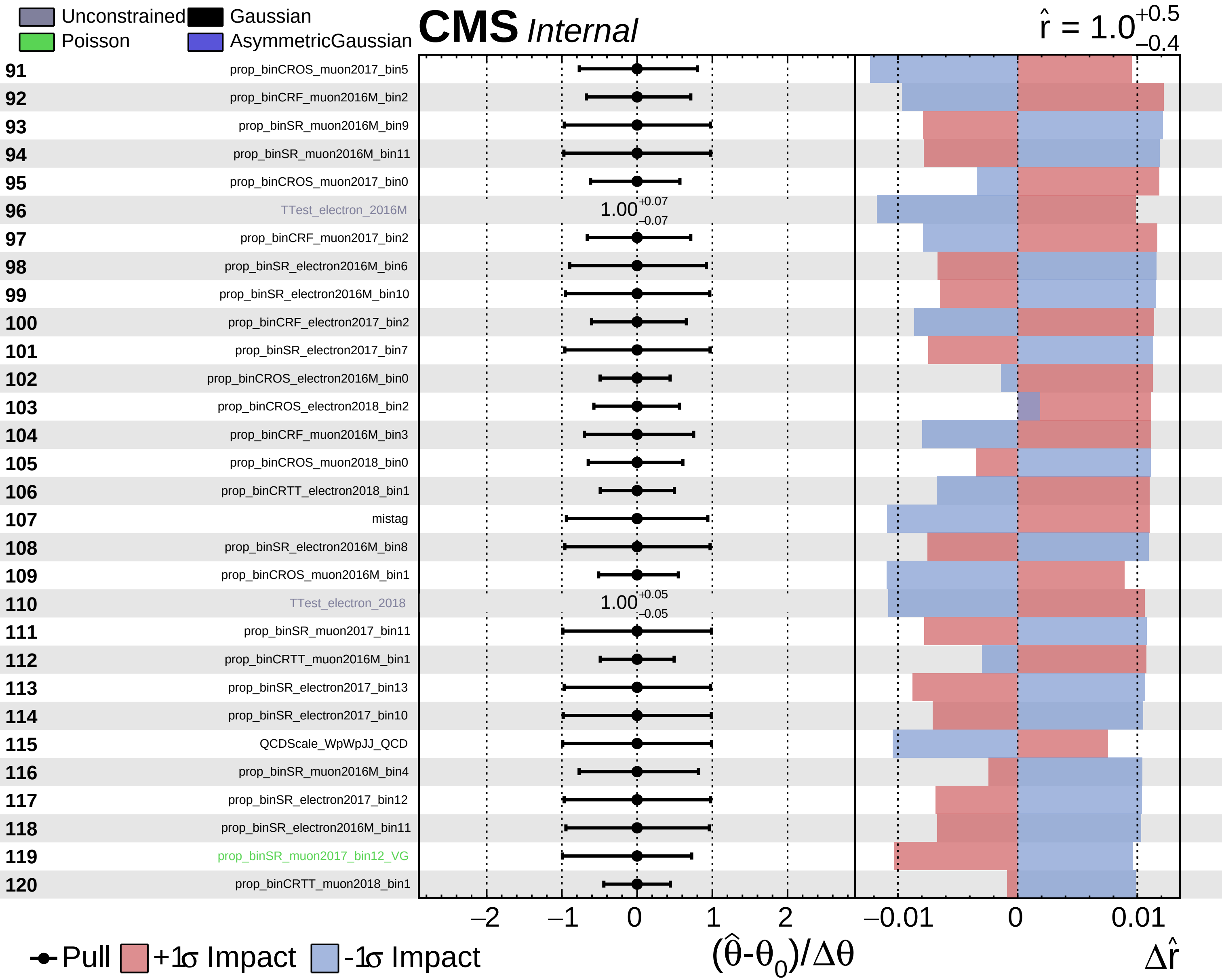
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
1 QCDScale_sig
2 prop_binCRF_muon2018_bin0
3 FR_sys_muon
4 prop_binCRF_muon2017_bin0
5 prop_binCRF_muon2016M_bin0
6 prop_binSR_muon2018_bin0
7 prop_binCRF_electron2017_bin0
8 prop_binCRF_electron2018_bin0
9 FR_sys_electron
10 prop_binCRF_electron2016M_bin0
11 OSest_muon_2018 $1.00^{+0.18}_{-0.18}$
12 prop_binSR_muon2017_bin0
13 OSest_muon_2017 $1.00^{+0.24}_{-0.24}$
14 prop_binCRF_muon2018_bin1
15 PF
16 OSest_muon_2016M $1.00^{+0.29}_{-0.29}$
17 prop_binSR_muon2017_bin7_WrongSign
18 prop_binSR_muon2016M_bin0
19 OSest_electron_2017 $1.00^{+0.29}_{-0.29}$
20 prop_binCRF_muon2016M_bin1
21 OSest_electron_2018 $1.00^{+0.29}_{-0.29}$
22 OSest_electron_2016M $1.00^{+0.30}_{-0.30}$
23 prop_binSR_muon2018_bin6
24 prop_binSR_muon2018_bin8
25 prop_binSR_electron2018_bin0
26 prop_binSR_muon2018_bin12
27 prop_binCRF_electron2018_bin1
28 prop_binSR_muon2018_bin7
29 prop_binCRF_muon2017_bin1
30 prop_binSR_muon2018_bin10
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



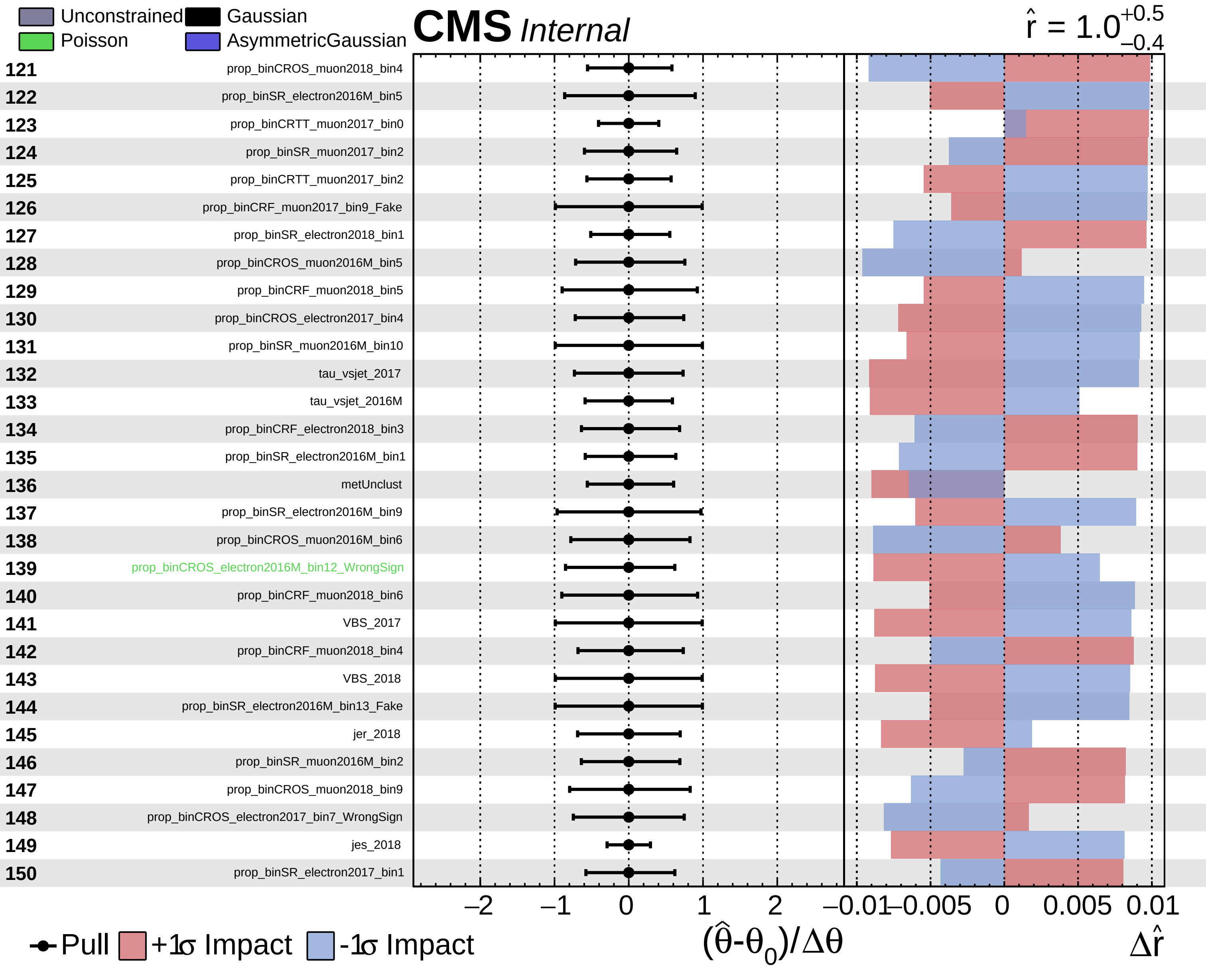
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
31 prop_binSR_muon2018_bin9
32 prop_binCRF_electron2016M_bin1
33 prop_binSR_electron2016M_bin0
34 prop_binSR_muon2018_bin5
35 prop_binCRTT_muon2016M_bin0
36 prop_binSR_electron2017_bin0
37 prop_binSR_muon2018_bin4
38 prop_binSR_electron2018_bin8
39 prop_binSR_electron2018_bin7
40 prop_binSR_electron2018_bin9
41 prop_binSR_electron2018_bin6
42 prop_binSR_muon2017_bin6
43 prop_binSR_muon2018_bin13
44 prop_binCRF_muon2018_bin2
45 prop_binSR_electron2018_bin11
46 prop_binCROS_electron2018_bin0
47 prop_binSR_muon2018_bin1
48 prop_binSR_muon2017_bin5
49 prop_binCRF_electron2017_bin1
50 prop_binCROS_electron2017_bin1
51 prop_binSR_muon2016M_bin7
52 prop_binSR_electron2018_bin5
53 pdf_Tot
54 prop_binSR_muon2016M_bin8
55 prop_binSR_muon2017_bin8
56 prop_binSR_muon2016M_bin1
57 pu
58 prop_binCRF_electron2018_bin2
59 prop_binSR_muon2017_bin4
60 prop_binSR_muon2018_bin11
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



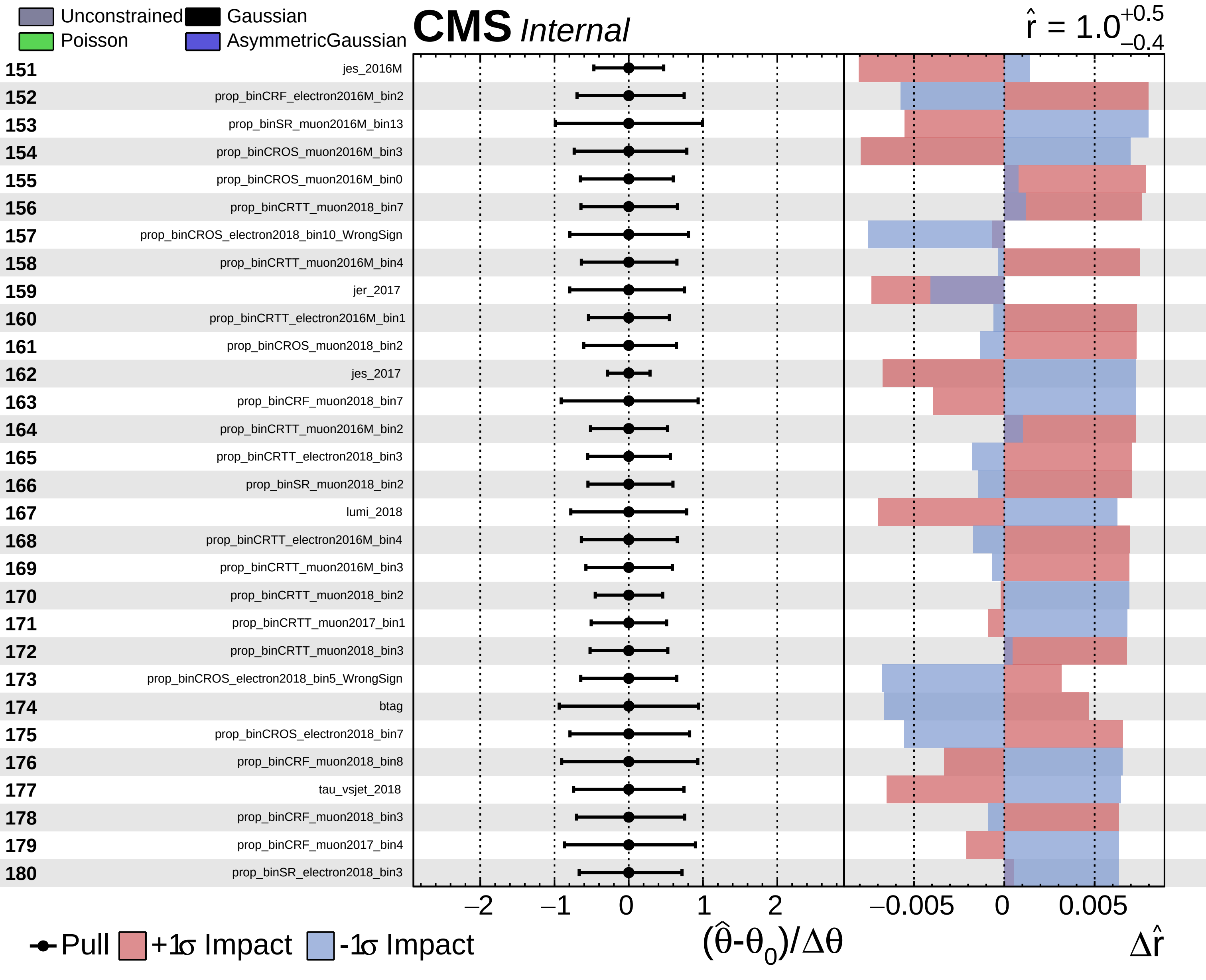
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.5 −0.4
61 prop_binSR_muon2016M_bin6
62 prop_binSR_muon2016M_bin5
63 prop_binCROS_muon2018_bin3
64 prop_binSR_muon2017_bin1
65 prop_binCROS_muon2016M_bin2
66 prop_binSR_electron2017_bin9
67 FSR
68 prop_binSR_electron2017_bin5
69 prop_binSR_electron2017_bin6
70 prop_binCROS_electron2018_bin6
71 prop_binSR_muon2017_bin3
72 prop_binSR_electron2018_bin10
73 prop_binCROS_electron2017_bin0
74 prop_binSR_electron2017_bin4
75 prop_binSR_muon2017_bin9
76 TTest_muon_2017 1.00 +0.05 −0.05
77 QCDScale_WrongSign
78 prop_binSR_electron2017_bin8
79 prop_binCROS_muon2017_bin7_WrongSign
80 prop_binSR_electron2016M_bin7
81 TTest_muon_2016M 1.00 +0.06 −0.06
82 prop_binSR_electron2018_bin4
83 prop_binCROS_electron2016M_bin4
84 prop_binCRTT_electron2016M_bin0
85 prop_binCRTT_electron2017_bin0
86 prop_binSR_electron2016M_bin4
87 TTest_muon_2018 1.00 +0.05 −0.05
88 TTest_electron_2017 1.00 +0.05 −0.05
89 prop_binCROS_muon2017_bin3
90 prop_binCROS_electron2016M_bin2
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

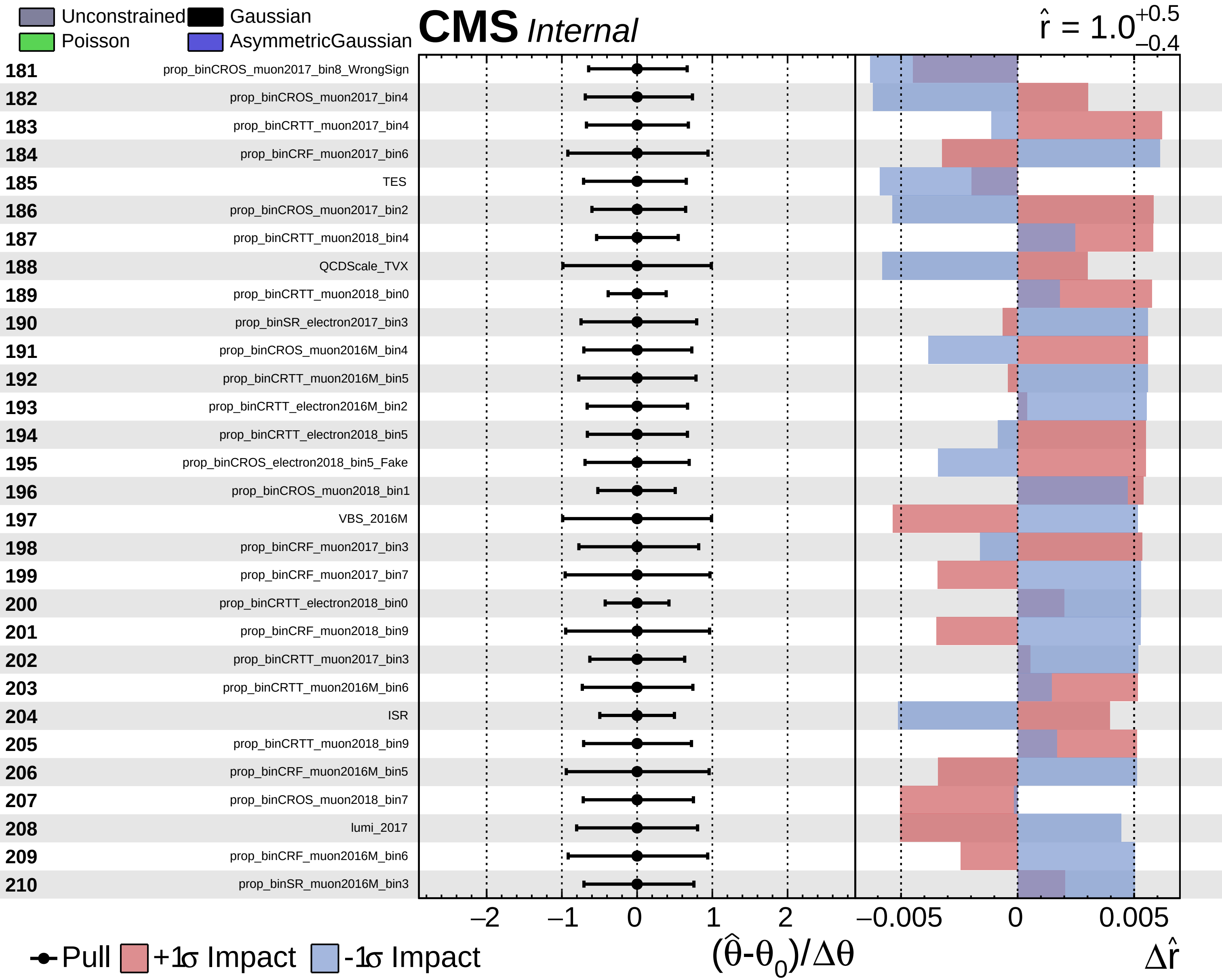


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
91 prop_binCROS_muon2017_bin5
92 prop_binCRF_muon2016M_bin2
93 prop_binSR_muon2016M_bin9
94 prop_binSR_muon2016M_bin11
95 prop_binCROS_muon2017_bin0
96 TTest_electron_2016M
$1.00^{+0.07}_{-0.07}$
97 prop_binCRF_muon2017_bin2
98 prop_binSR_electron2016M_bin6
99 prop_binSR_electron2016M_bin10
100 prop_binCRF_electron2017_bin2
101 prop_binSR_electron2017_bin7
102 prop_binCROS_electron2016M_bin0
103 prop_binCROS_electron2018_bin2
104 prop_binCRF_muon2016M_bin3
105 prop_binCROS_muon2018_bin0
106 prop_binCRTT_electron2018_bin1
107 mistag
108 prop_binSR_electron2016M_bin8
109 prop_binCROS_muon2016M_bin1
110 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
111 prop_binSR_muon2017_bin11
112 prop_binCRTT_muon2016M_bin1
113 prop_binSR_electron2017_bin13
114 prop_binSR_electron2017_bin10
115 QCDScale_WpWpJJ_QCD
116 prop_binSR_muon2016M_bin4
117 prop_binSR_electron2017_bin12
118 prop_binSR_electron2016M_bin11
119 prop_binSR_muon2017_bin12_VG
120 prop_binCRTT_muon2018_bin1
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

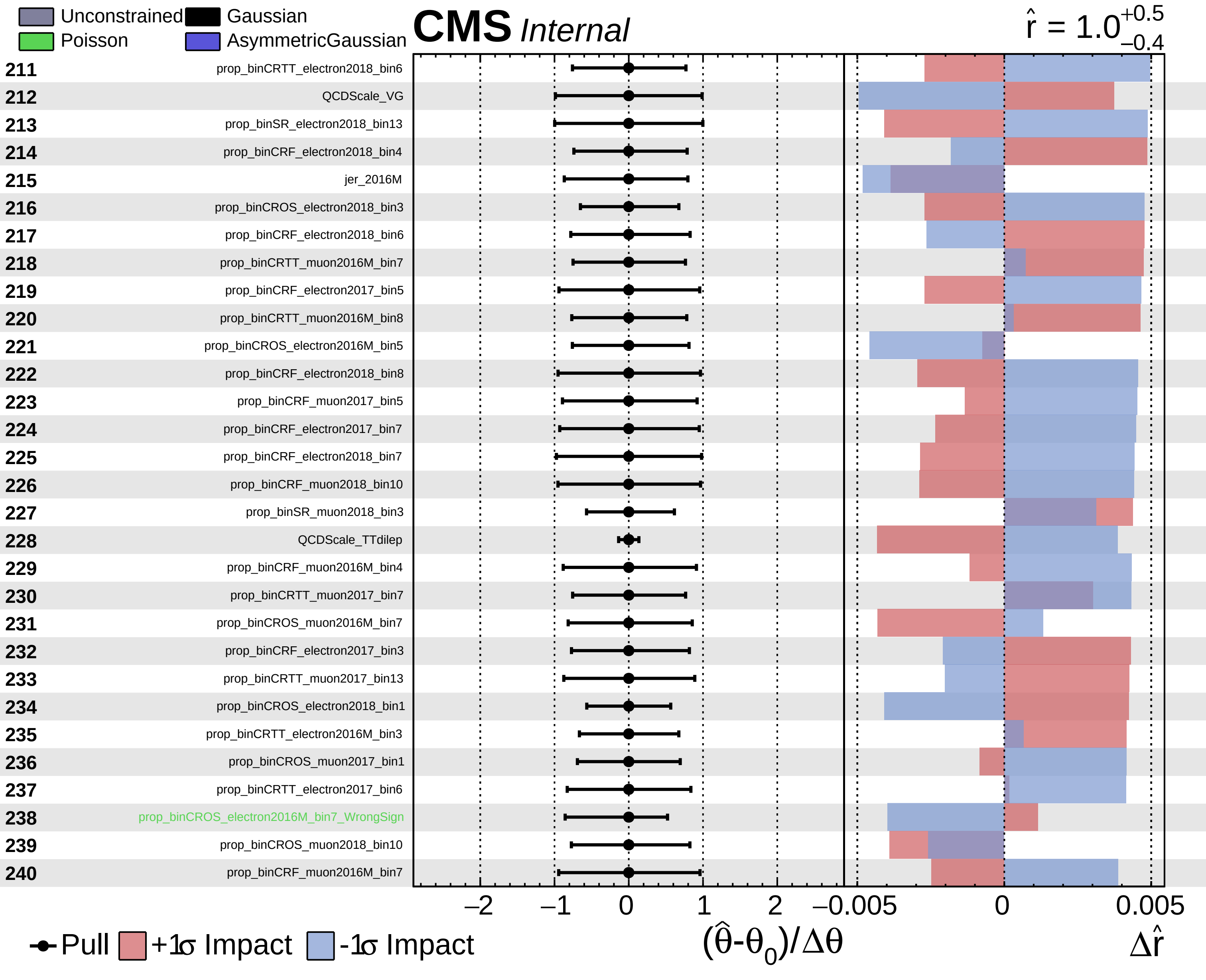


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
121 prop_binCROS_muon2018_bin4
122 prop_binSR_electron2016M_bin5
123 prop_binCRTT_muon2017_bin0
124 prop_binSR_muon2017_bin2
125 prop_binCRTT_muon2017_bin2
126 prop_binCRF_muon2017_bin9_Fake
127 prop_binSR_electron2018_bin1
128 prop_binCROS_muon2016M_bin5
129 prop_binCRF_muon2018_bin5
130 prop_binCROS_electron2017_bin4
131 prop_binSR_muon2016M_bin10
132 tau_vsjet_2017
133 tau_vsjet_2016M
134 prop_binCRF_electron2018_bin3
135 prop_binSR_electron2016M_bin1
136 metUnclust
137 prop_binSR_electron2016M_bin9
138 prop_binCROS_muon2016M_bin6
139 prop_binCROS_electron2016M_bin12_WrongSign
140 prop_binCRF_muon2018_bin6
141 VBS_2017
142 prop_binCRF_muon2018_bin4
143 VBS_2018
144 prop_binSR_electron2016M_bin13_Fake
145 jer_2018
146 prop_binSR_muon2016M_bin2
147 prop_binCROS_muon2018_bin9
148 prop_binCROS_electron2017_bin7_WrongSign
149 jes_2018
150 prop_binSR_electron2017_bin1
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

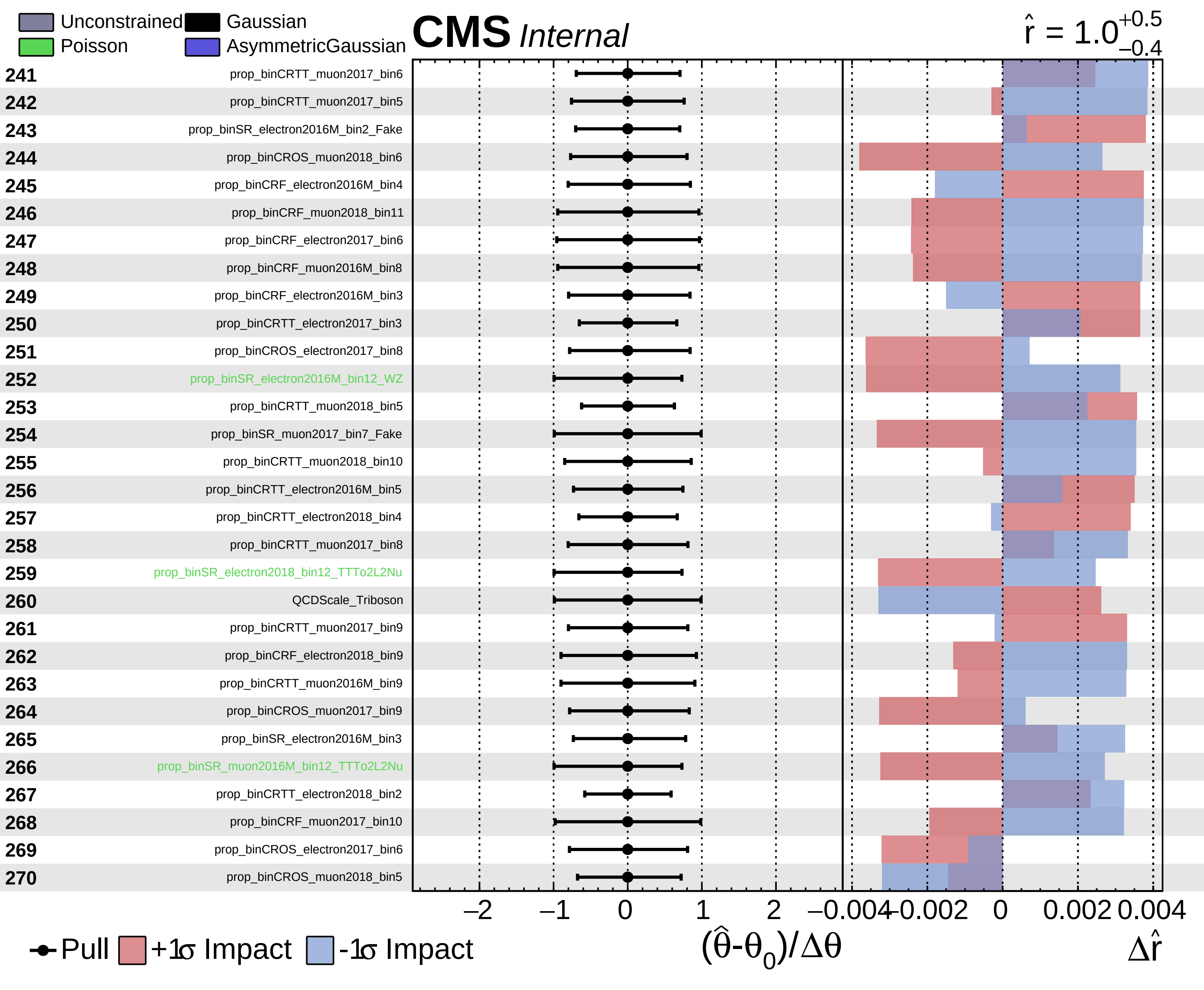

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
151 jes_2016M
152 prop_binCRF_electron2016M_bin2
153 prop_binSR_muon2016M_bin13
154 prop_binCROS_muon2016M_bin3
155 prop_binCROS_muon2016M_bin0
156 prop_binCRTT_muon2018_bin7
157 prop_binCROS_electron2018_bin10_WrongSign
158 prop_binCRTT_muon2016M_bin4
159 jer_2017
160 prop_binCRTT_electron2016M_bin1
161 prop_binCROS_muon2018_bin2
162 jes_2017
163 prop_binCRF_muon2018_bin7
164 prop_binCRTT_muon2016M_bin2
165 prop_binCRTT_electron2018_bin3
166 prop_binSR_muon2018_bin2
167 lumi_2018
168 prop_binCRTT_electron2016M_bin4
169 prop_binCRTT_muon2016M_bin3
170 prop_binCRTT_muon2018_bin2
171 prop_binCRTT_muon2017_bin1
172 prop_binCRTT_muon2018_bin3
173 prop_binCROS_electron2018_bin5_WrongSign
174 btag
175 prop_binCROS_electron2018_bin7
176 prop_binCRF_muon2018_bin8
177 tau_vsjet_2018
178 prop_binCRF_muon2018_bin3
179 prop_binCRF_muon2017_bin4
180 prop_binSR_electron2018_bin3
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



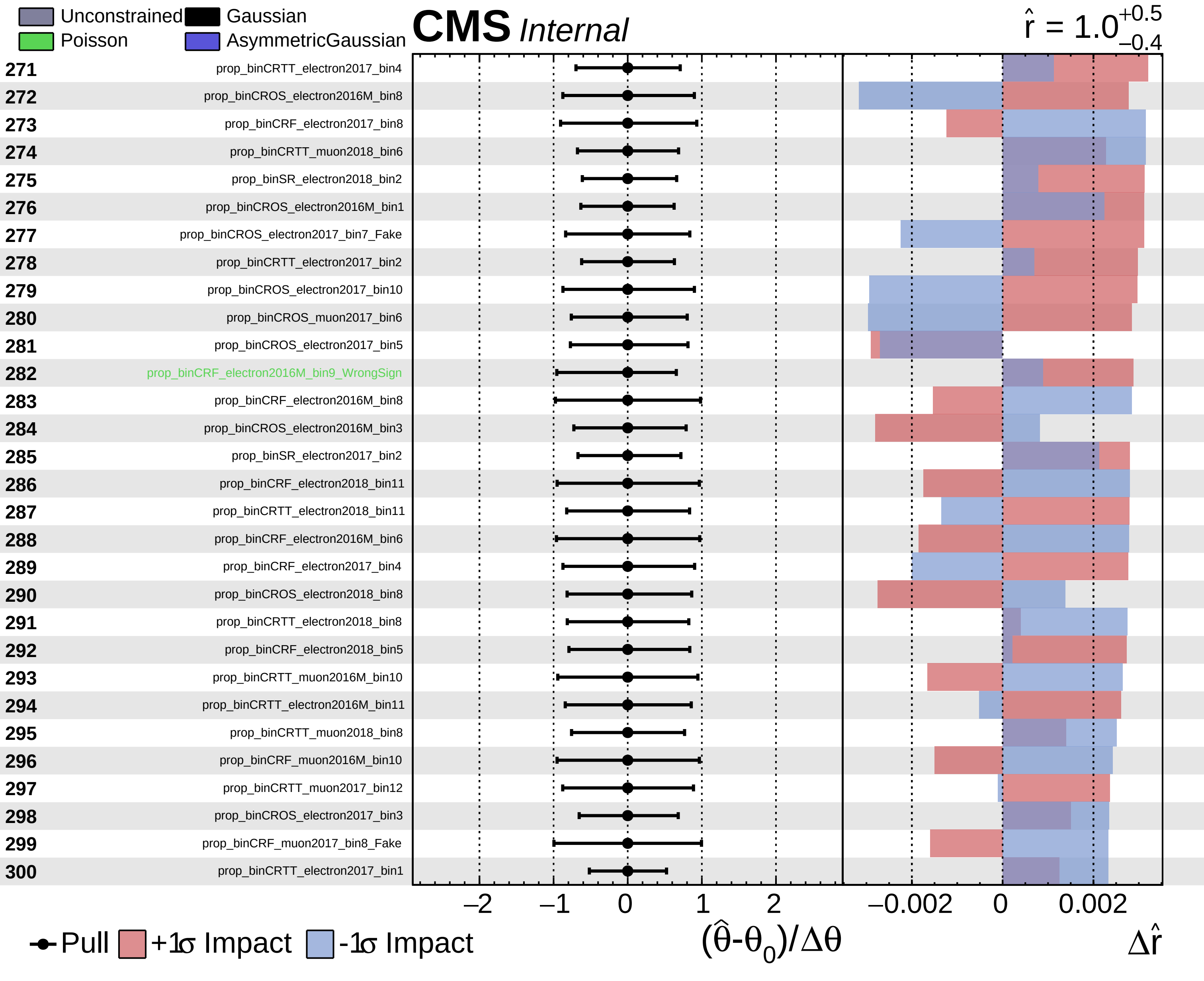
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
181 prop_binCROS_muon2017_bin8_WrongSign
182 prop_binCROS_muon2017_bin4
183 prop_binCRTT_muon2017_bin4
184 prop_binCRF_muon2017_bin6
185 TES
186 prop_binCROS_muon2017_bin2
187 prop_binCRTT_muon2018_bin4
188 QCDScale_TVX
189 prop_binCRTT_muon2018_bin0
190 prop_binSR_electron2017_bin3
191 prop_binCROS_muon2016M_bin4
192 prop_binCRTT_muon2016M_bin5
193 prop_binCRTT_electron2016M_bin2
194 prop_binCRTT_electron2018_bin5
195 prop_binCROS_electron2018_bin5_Fake
196 prop_binCROS_muon2018_bin1
197 VBS_2016M
198 prop_binCRF_muon2017_bin3
199 prop_binCRF_muon2017_bin7
200 prop_binCRTT_electron2018_bin0
201 prop_binCRF_muon2018_bin9
202 prop_binCRTT_muon2017_bin3
203 prop_binCRTT_muon2016M_bin6
204 ISR
205 prop_binCRTT_muon2018_bin9
206 prop_binCRF_muon2016M_bin5
207 prop_binCROS_muon2018_bin7
208 lumi_2017
209 prop_binCRF_muon2016M_bin6
210 prop_binSR_muon2016M_bin3
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



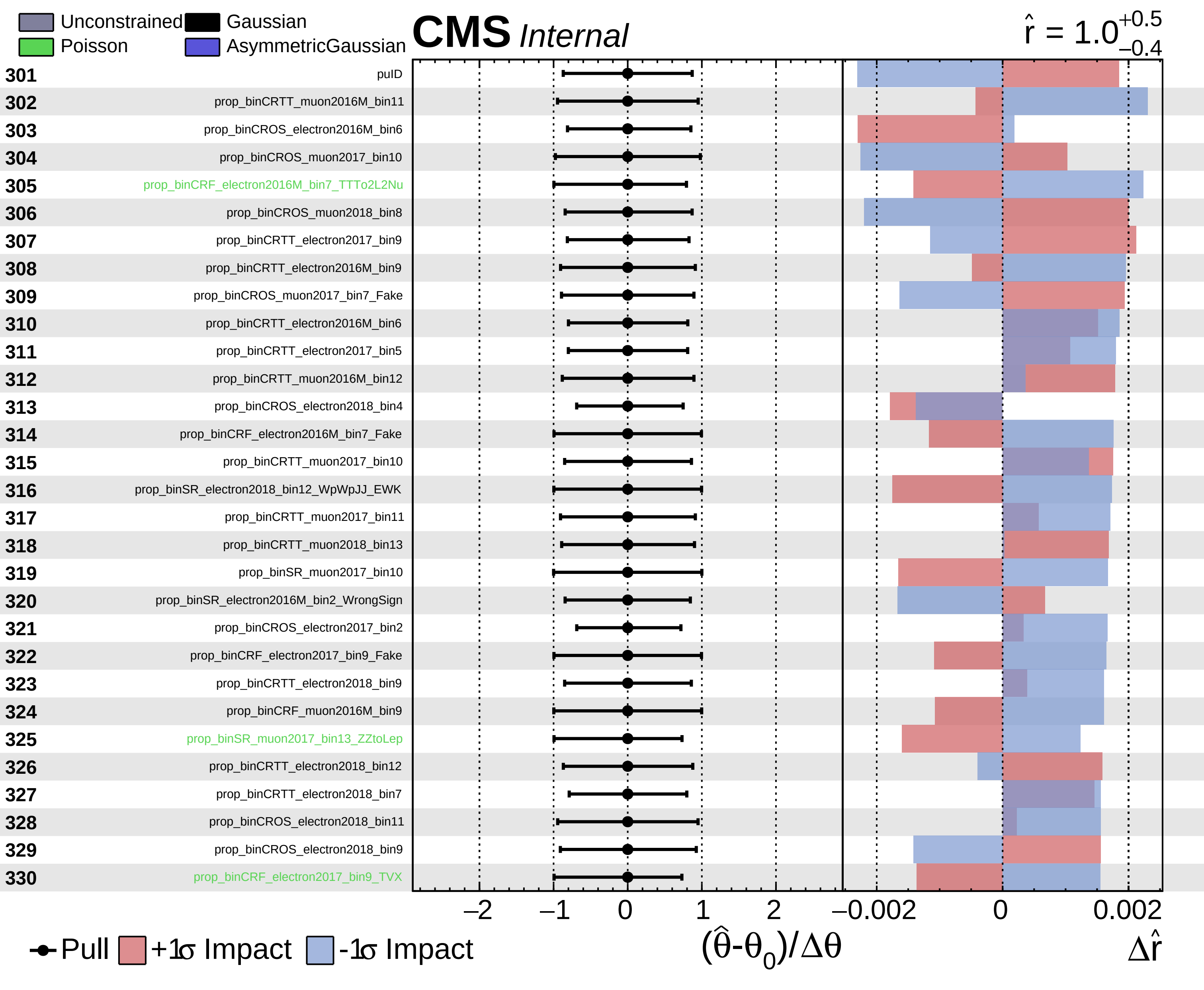
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
211 prop_binCRTT_electron2018_bin6
212 QCDScale_VG
213 prop_binSR_electron2018_bin13
214 prop_binCRF_electron2018_bin4
215 jer_2016M
216 prop_binCROS_electron2018_bin3
217 prop_binCRF_electron2018_bin6
218 prop_binCRTT_muon2016M_bin7
219 prop_binCRF_electron2017_bin5
220 prop_binCRTT_muon2016M_bin8
221 prop_binCROS_electron2016M_bin5
222 prop_binCRF_electron2018_bin8
223 prop_binCRF_muon2017_bin5
224 prop_binCRF_electron2017_bin7
225 prop_binCRF_electron2018_bin7
226 prop_binCRF_muon2018_bin10
227 prop_binSR_muon2018_bin3
228 QCDScale_TTdilep
229 prop_binCRF_muon2016M_bin4
230 prop_binCRTT_muon2017_bin7
231 prop_binCROS_muon2016M_bin7
232 prop_binCRF_electron2017_bin3
233 prop_binCRTT_muon2017_bin13
234 prop_binCROS_electron2018_bin1
235 prop_binCRTT_electron2016M_bin3
236 prop_binCROS_muon2017_bin1
237 prop_binCRTT_electron2017_bin6
238 prop_binCROS_electron2016M_bin7_WrongSign
239 prop_binCROS_muon2018_bin10
240 prop_binCRF_muon2016M_bin7
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



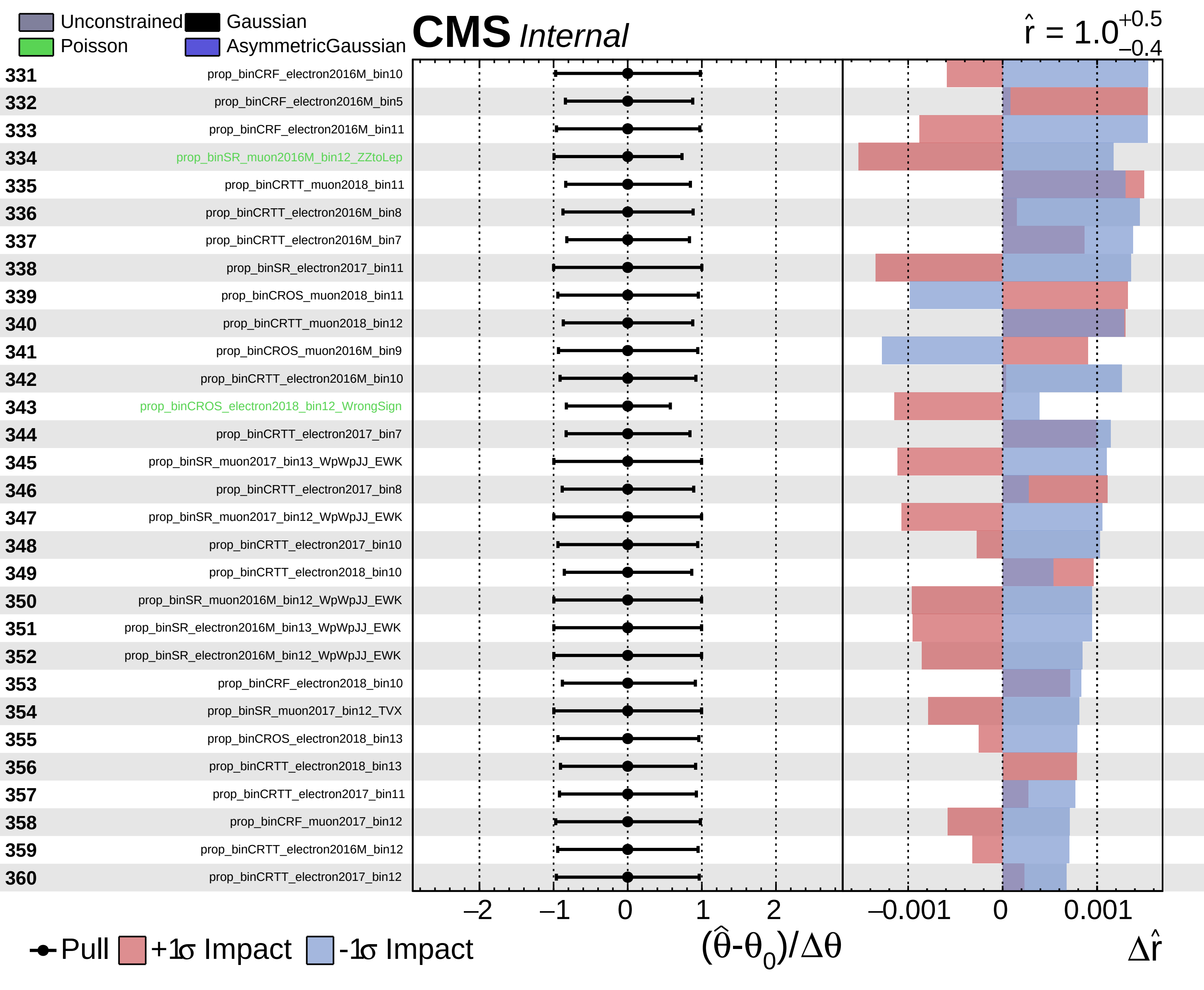
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
241 prop_binCRTT_muon2017_bin6
242 prop_binCRTT_muon2017_bin5
243 prop_binSR_electron2016M_bin2_Fake
244 prop_binCROS_muon2018_bin6
245 prop_binCRF_electron2016M_bin4
246 prop_binCRF_muon2018_bin11
247 prop_binCRF_electron2017_bin6
248 prop_binCRF_muon2016M_bin8
249 prop_binCRF_electron2016M_bin3
250 prop_binCRTT_electron2017_bin3
251 prop_binCROS_electron2017_bin8
252 prop_binSR_electron2016M_bin12_WZ
253 prop_binCRTT_muon2018_bin5
254 prop_binSR_muon2017_bin7_Fake
255 prop_binCRTT_muon2018_bin10
256 prop_binCRTT_electron2016M_bin5
257 prop_binCRTT_electron2018_bin4
258 prop_binCRTT_muon2017_bin8
259 prop_binSR_electron2018_bin12_TTTo2L2Nu
260 QCDScale_Triboson
261 prop_binCRTT_muon2017_bin9
262 prop_binCRF_electron2018_bin9
263 prop_binCRTT_muon2016M_bin9
264 prop_binCROS_muon2017_bin9
265 prop_binSR_electron2016M_bin3
266 prop_binSR_muon2016M_bin12_TTTo2L2Nu
267 prop_binCRTT_electron2018_bin2
268 prop_binCRF_muon2017_bin10
269 prop_binCROS_electron2017_bin6
270 prop_binCROS_muon2018_bin5
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



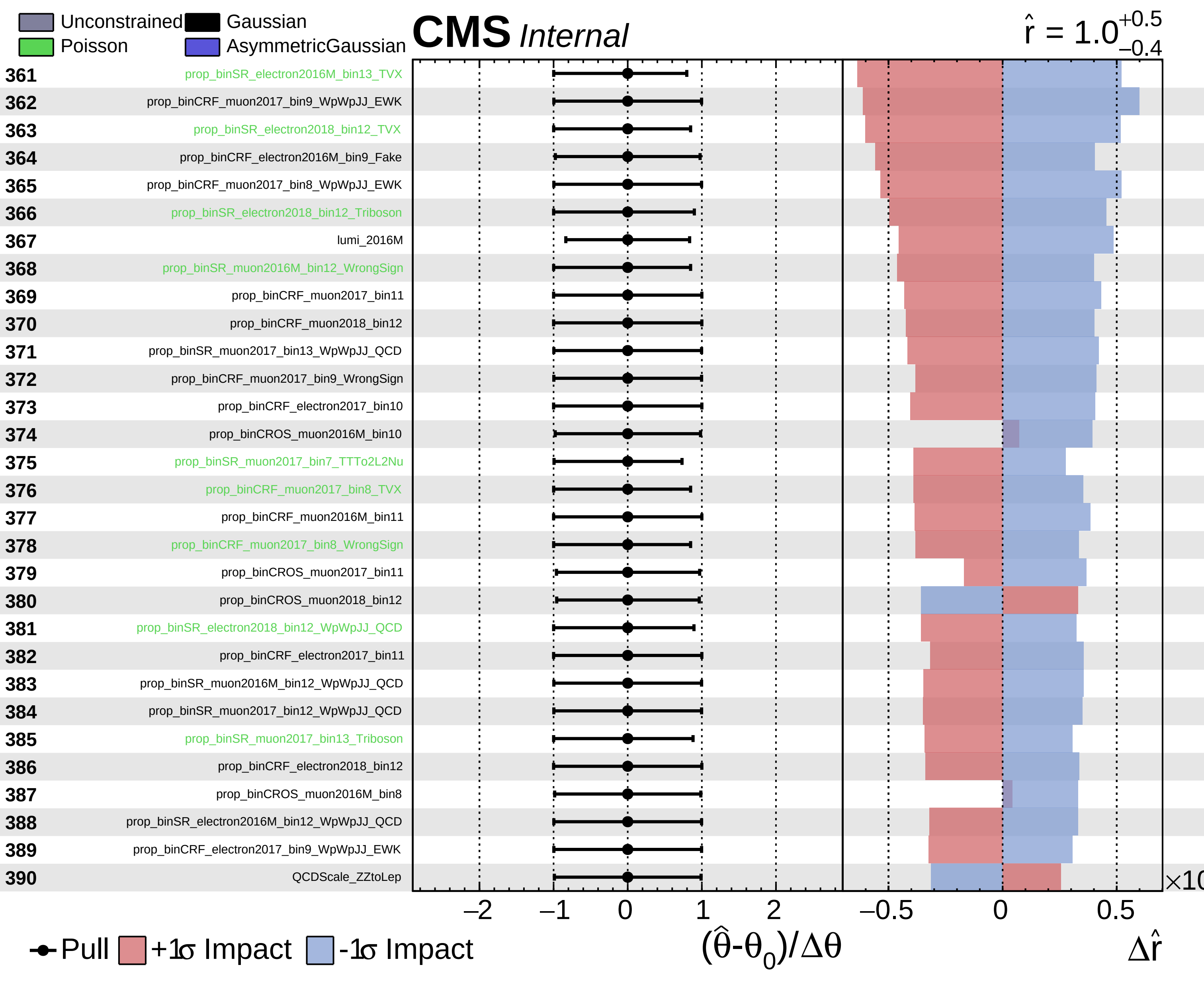
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
271 prop_binCRTT_electron2017_bin4
272 prop_binCROS_electron2016M_bin8
273 prop_binCRF_electron2017_bin8
274 prop_binCRTT_muon2018_bin6
275 prop_binSR_electron2018_bin2
276 prop_binCROS_electron2016M_bin1
277 prop_binCROS_electron2017_bin7_Fake
278 prop_binCRTT_electron2017_bin2
279 prop_binCROS_electron2017_bin10
280 prop_binCROS_muon2017_bin6
281 prop_binCROS_electron2017_bin5
282 prop_binCRF_electron2016M_bin9_WrongSign
283 prop_binCRF_electron2016M_bin8
284 prop_binCROS_electron2016M_bin3
285 prop_binSR_electron2017_bin2
286 prop_binCRF_electron2018_bin11
287 prop_binCRTT_electron2018_bin11
288 prop_binCRF_electron2016M_bin6
289 prop_binCRF_electron2017_bin4
290 prop_binCROS_electron2018_bin8
291 prop_binCRTT_electron2018_bin8
292 prop_binCRF_electron2018_bin5
293 prop_binCRTT_muon2016M_bin10
294 prop_binCRTT_electron2016M_bin11
295 prop_binCRTT_muon2018_bin8
296 prop_binCRF_muon2016M_bin10
297 prop_binCRTT_muon2017_bin12
298 prop_binCROS_electron2017_bin3
299 prop_binCRF_muon2017_bin8_Fake
300 prop_binCRTT_electron2017_bin1
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

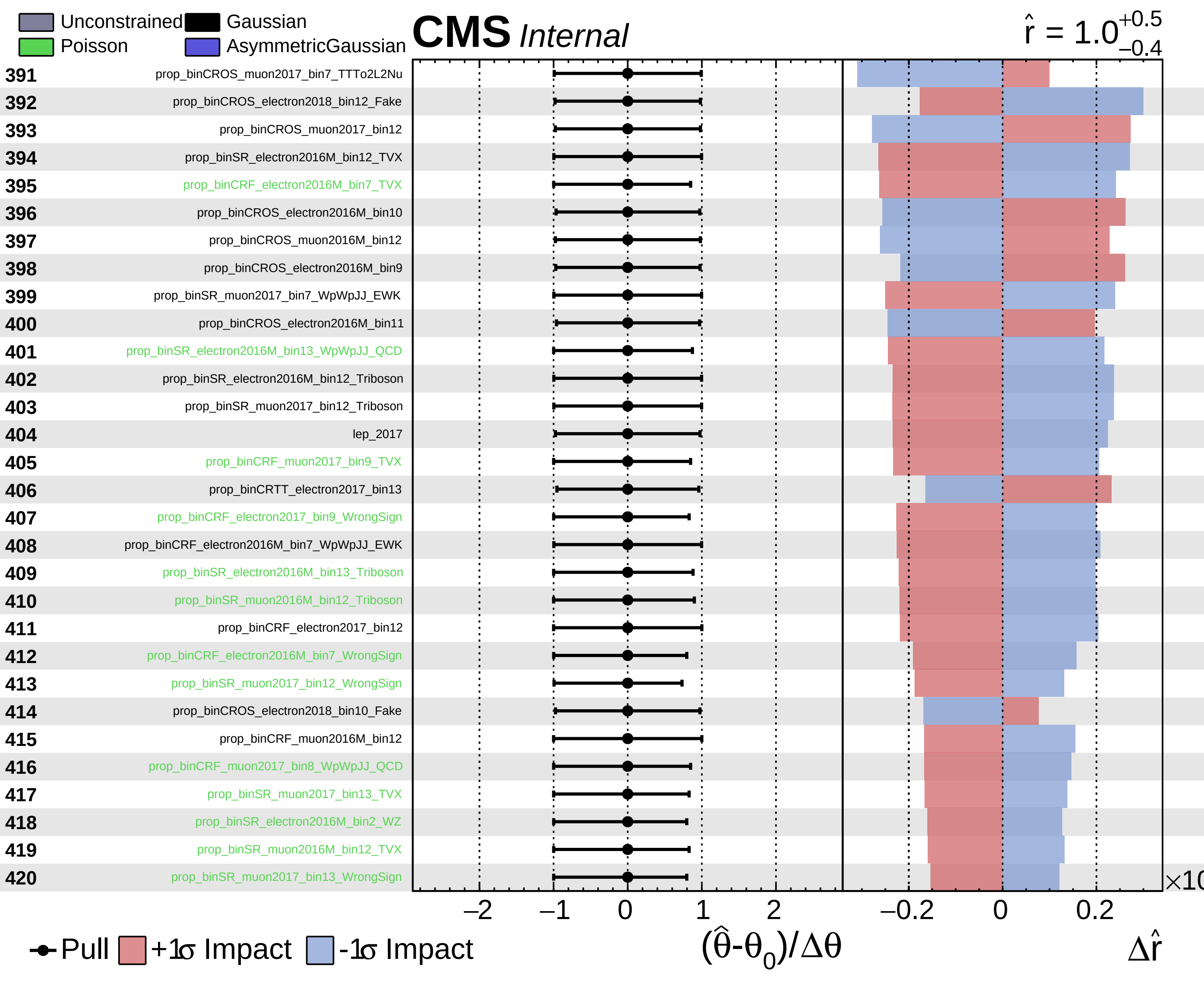


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
301 puID
302 prop_binCRTT_muon2016M_bin11
303 prop_binCROS_electron2016M_bin6
304 prop_binCROS_muon2017_bin10
305 prop_binCRF_electron2016M_bin7_TTTo2L2Nu
306 prop_binCROS_muon2018_bin8
307 prop_binCRTT_electron2017_bin9
308 prop_binCRTT_electron2016M_bin9
309 prop_binCROS_muon2017_bin7_Fake
310 prop_binCRTT_electron2016M_bin6
311 prop_binCRTT_electron2017_bin5
312 prop_binCRTT_muon2016M_bin12
313 prop_binCROS_electron2018_bin4
314 prop_binCRF_electron2016M_bin7_Fake
315 prop_binCRTT_muon2017_bin10
316 prop_binSR_electron2018_bin12_WpWpJJ_EWK
317 prop_binCRTT_muon2017_bin11
318 prop_binCRTT_muon2018_bin13
319 prop_binSR_muon2017_bin10
320 prop_binSR_electron2016M_bin2_WrongSign
321 prop_binCROS_electron2017_bin2
322 prop_binCRF_electron2017_bin9_Fake
323 prop_binCRTT_electron2018_bin9
324 prop_binCRF_muon2016M_bin9
325 prop_binSR_muon2017_bin13_ZZtoLep
326 prop_binCRTT_electron2018_bin12
327 prop_binCRTT_electron2018_bin7
328 prop_binCROS_electron2018_bin11
329 prop_binCROS_electron2018_bin9
330 prop_binCRF_electron2017_bin9_TVX
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

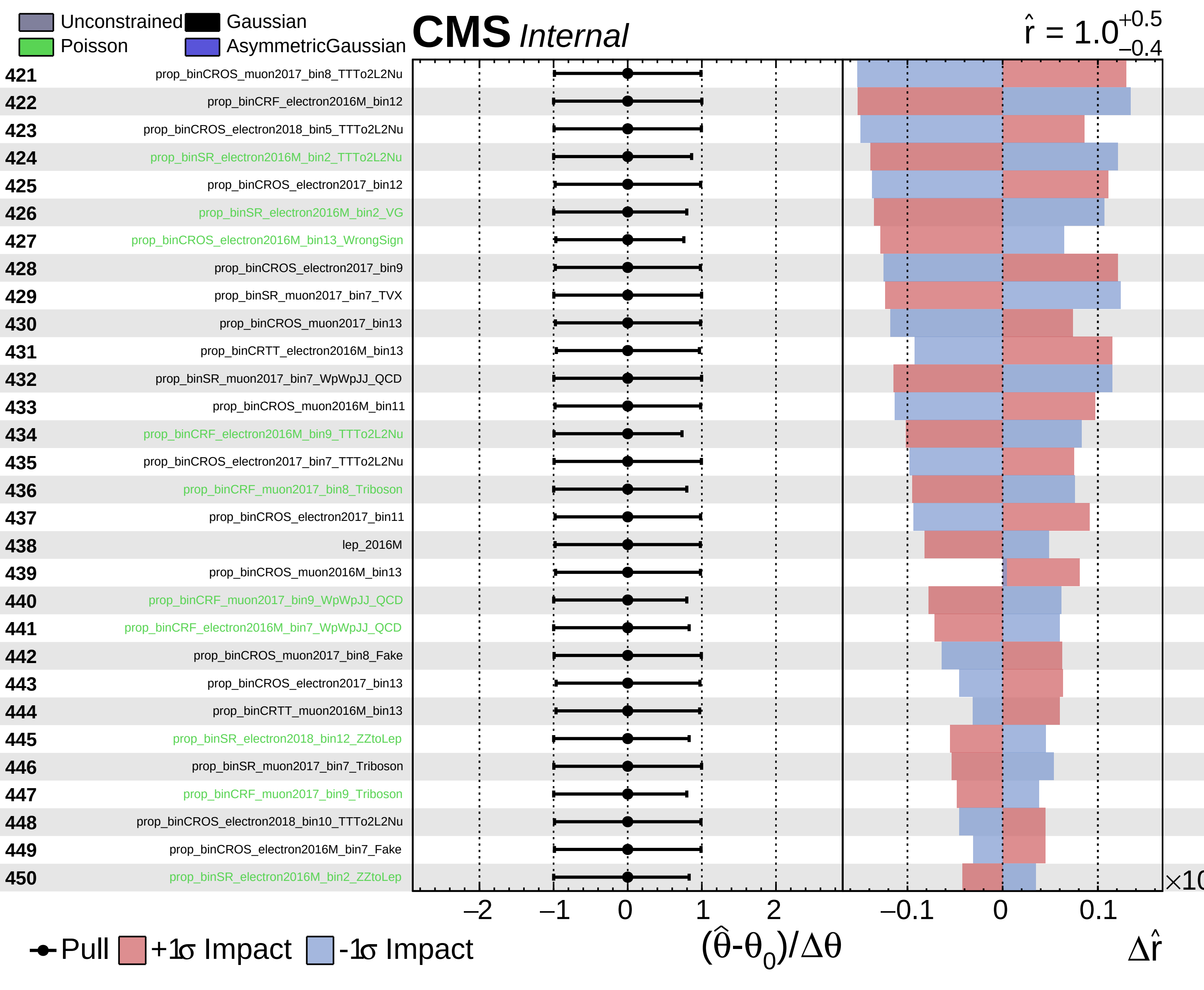


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
331 prop_binCRF_electron2016M_bin10
332 prop_binCRF_electron2016M_bin5
333 prop_binCRF_electron2016M_bin11
334 prop_binSR_muon2016M_bin12_ZZtoLep
335 prop_binCRTT_muon2018_bin11
336 prop_binCRTT_electron2016M_bin8
337 prop_binCRTT_electron2016M_bin7
338 prop_binSR_electron2017_bin11
339 prop_binCROS_muon2018_bin11
340 prop_binCRTT_muon2018_bin12
341 prop_binCROS_muon2016M_bin9
342 prop_binCRTT_electron2016M_bin10
343 prop_binCROS_electron2018_bin12_WrongSign
344 prop_binCRTT_electron2017_bin7
345 prop_binSR_muon2017_bin13_WpWpJJ_EWK
346 prop_binCRTT_electron2017_bin8
347 prop_binSR_muon2017_bin12_WpWpJJ_EWK
348 prop_binCRTT_electron2017_bin10
349 prop_binCRTT_electron2018_bin10
350 prop_binSR_muon2016M_bin12_WpWpJJ_EWK
351 prop_binSR_electron2016M_bin13_WpWpJJ_EWK
352 prop_binSR_electron2016M_bin12_WpWpJJ_EWK
353 prop_binCRF_electron2018_bin10
354 prop_binSR_muon2017_bin12_TVX
355 prop_binCROS_electron2018_bin13
356 prop_binCRTT_electron2018_bin13
357 prop_binCRTT_electron2017_bin11
358 prop_binCRF_muon2017_bin12
359 prop_binCRTT_electron2016M_bin12
360 prop_binCRTT_electron2017_bin12
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

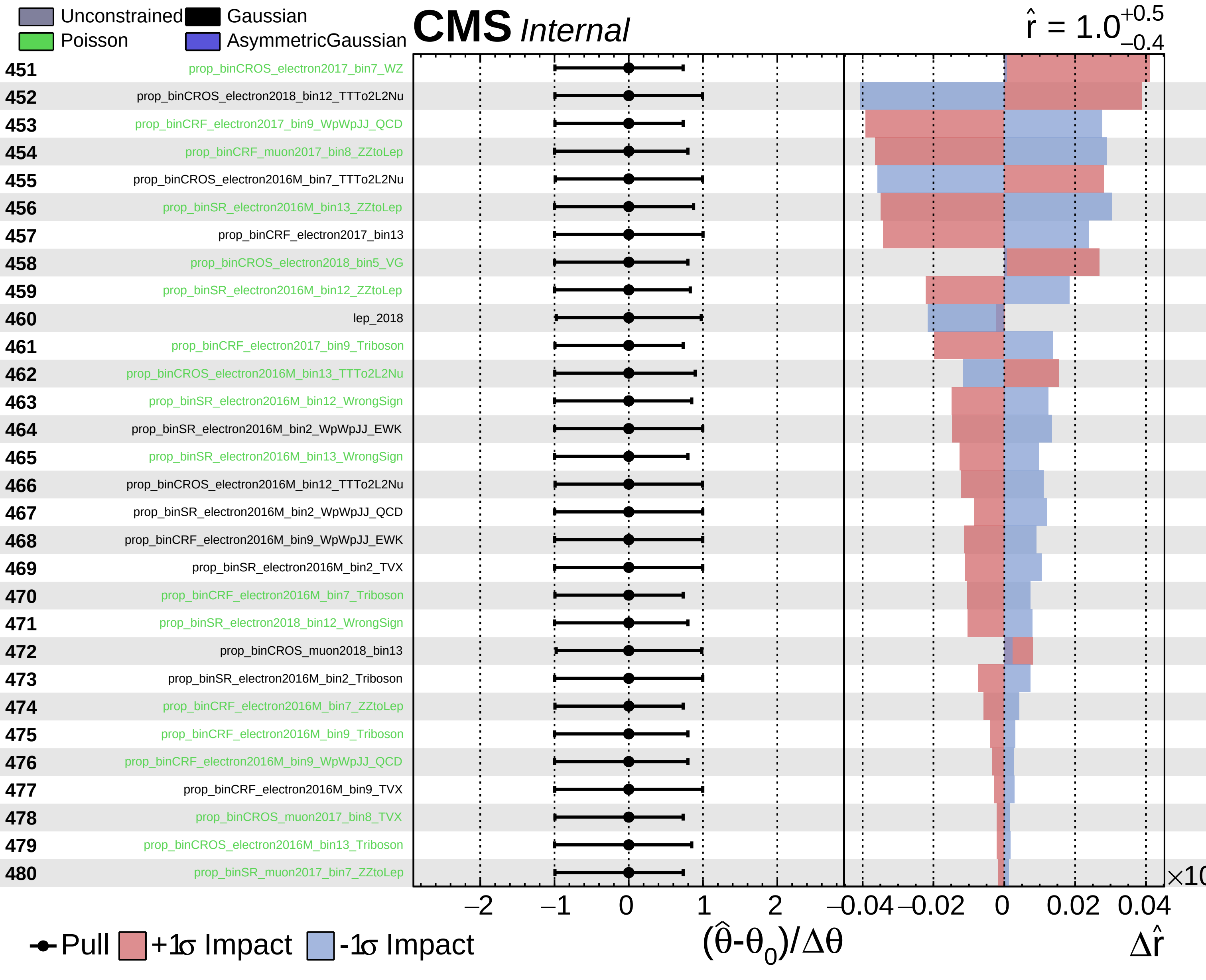

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
361 prop_binSR_electron2016M_bin13_TVX
362 prop_binCRF_muon2017_bin9_WpWpJJ_EWK
363 prop_binSR_electron2018_bin12_TVX
364 prop_binCRF_electron2016M_bin9_Fake
365 prop_binCRF_muon2017_bin8_WpWpJJ_EWK
366 prop_binSR_electron2018_bin12_Triboson
367 lumi_2016M
368 prop_binSR_muon2016M_bin12_WrongSign
369 prop_binCRF_muon2017_bin11
370 prop_binCRF_muon2018_bin12
371 prop_binSR_muon2017_bin13_WpWpJJ_QCD
372 prop_binCRF_muon2017_bin9_WrongSign
373 prop_binCRF_electron2017_bin10
374 prop_binCROS_muon2016M_bin10
375 prop_binSR_muon2017_bin7_TTTo2L2Nu
376 prop_binCRF_muon2017_bin8_TVX
377 prop_binCRF_muon2016M_bin11
378 prop_binCRF_muon2017_bin8_WrongSign
379 prop_binCROS_muon2017_bin11
380 prop_binCROS_muon2018_bin12
381 prop_binSR_electron2018_bin12_WpWpJJ_QCD
382 prop_binCRF_electron2017_bin11
383 prop_binSR_muon2016M_bin12_WpWpJJ_QCD
384 prop_binSR_muon2017_bin12_WpWpJJ_QCD
385 prop_binSR_muon2017_bin13_Triboson
386 prop_binCRF_electron2018_bin12
387 prop_binCROS_muon2016M_bin8
388 prop_binSR_electron2016M_bin12_WpWpJJ_QCD
389 prop_binCRF_electron2017_bin9_WpWpJJ_EWK
390 QCDScale_ZZtoLep
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

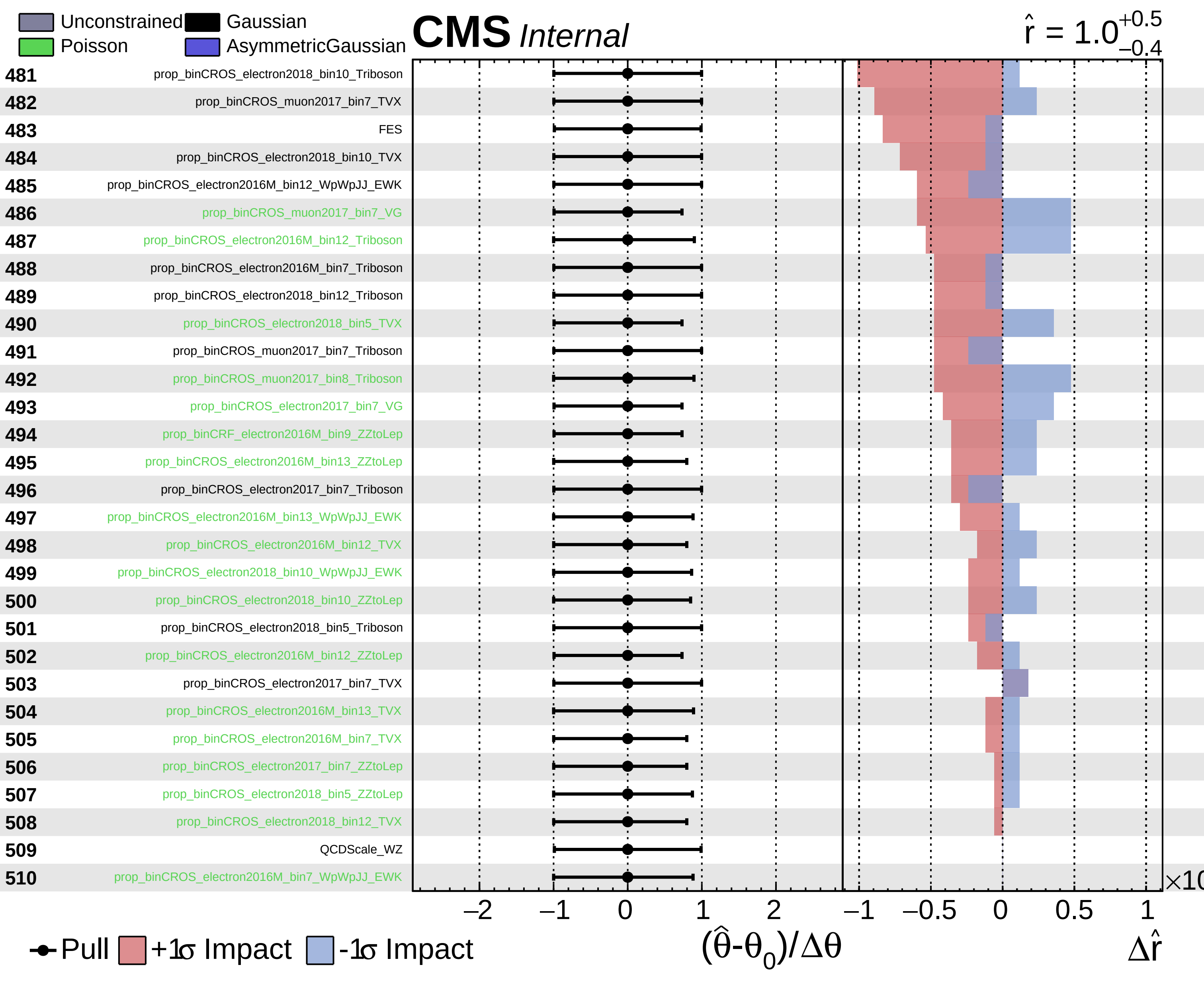


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
391 prop_binCROS_muon2017_bin7_TTTo2L2Nu
392 prop_binCROS_electron2018_bin12_Fake
393 prop_binCROS_muon2017_bin12
394 prop_binSR_electron2016M_bin12_TVX
395 prop_binCRF_electron2016M_bin7_TVX
396 prop_binCROS_electron2016M_bin10
397 prop_binCROS_muon2016M_bin12
398 prop_binCROS_electron2016M_bin9
399 prop_binSR_muon2017_bin7_WpWpJJ_EWK
400 prop_binCROS_electron2016M_bin11
401 prop_binSR_electron2016M_bin13_WpWpJJ_QCD
402 prop_binSR_electron2016M_bin12_Triboson
403 prop_binSR_muon2017_bin12_Triboson
404 lep_2017
405 prop_binCRF_muon2017_bin9_TVX
406 prop_binCRTT_electron2017_bin13
407 prop_binCRF_electron2017_bin9_WrongSign
408 prop_binCRF_electron2016M_bin7_WpWpJJ_EWK
409 prop_binSR_electron2016M_bin13_Triboson
410 prop_binSR_muon2016M_bin12_Triboson
411 prop_binCRF_electron2017_bin12
412 prop_binCRF_electron2016M_bin7_WrongSign
413 prop_binSR_muon2017_bin12_WrongSign
414 prop_binCROS_electron2018_bin10_Fake
415 prop_binCRF_muon2016M_bin12
416 prop_binCRF_muon2017_bin8_WpWpJJ_QCD
417 prop_binSR_muon2017_bin13_TVX
418 prop_binSR_electron2016M_bin2_WZ
419 prop_binSR_muon2016M_bin12_TVX
420 prop_binSR_muon2017_bin13_WrongSign
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

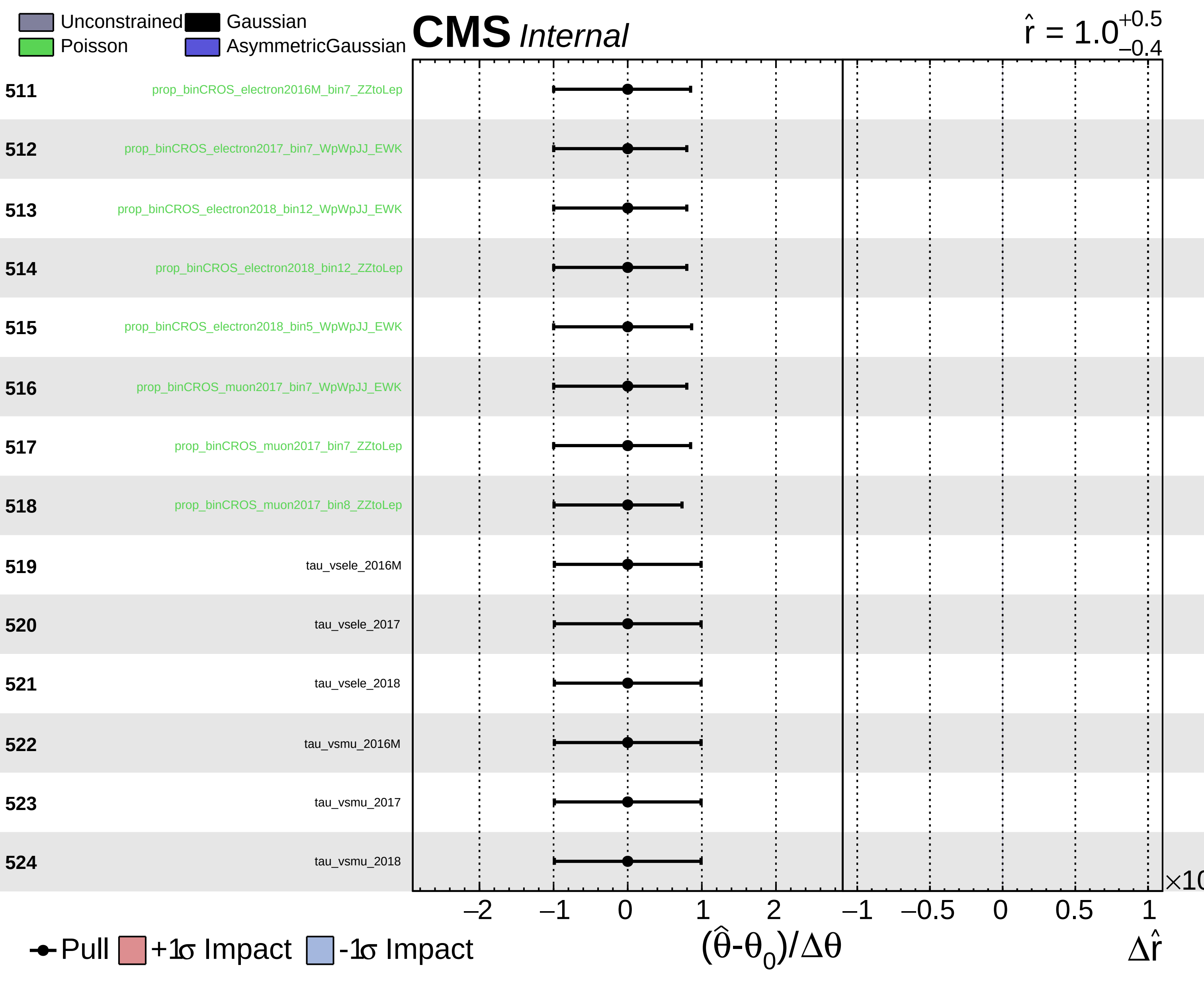


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
421 prop_binCROS_muon2017_bin8_TTTo2L2Nu
422 prop_binCRF_electron2016M_bin12
423 prop_binCROS_electron2018_bin5_TTTo2L2Nu
424 prop_binSR_electron2016M_bin2_TTTo2L2Nu
425 prop_binCROS_electron2017_bin12
426 prop_binSR_electron2016M_bin2_VG
427 prop_binCROS_electron2016M_bin13_WrongSign
428 prop_binCROS_electron2017_bin9
429 prop_binSR_muon2017_bin7_TVX
430 prop_binCROS_muon2017_bin13
431 prop_binCRTT_electron2016M_bin13
432 prop_binSR_muon2017_bin7_WpWpJJ_QCD
433 prop_binCROS_muon2016M_bin11
434 prop_binCRF_electron2016M_bin9_TTTo2L2Nu
435 prop_binCROS_electron2017_bin7_TTTo2L2Nu
436 prop_binCRF_muon2017_bin8_Triboson
437 prop_binCROS_electron2017_bin11
438 lep_2016M
439 prop_binCROS_muon2016M_bin13
440 prop_binCRF_muon2017_bin9_WpWpJJ_QCD
441 prop_binCRF_electron2016M_bin7_WpWpJJ_QCD
442 prop_binCROS_muon2017_bin8_Fake
443 prop_binCROS_electron2017_bin13
444 prop_binCRTT_muon2016M_bin13
445 prop_binSR_electron2018_bin12_ZZtoLep
446 prop_binSR_muon2017_bin7_Triboson
447 prop_binCRF_muon2017_bin9_Triboson
448 prop_binCROS_electron2018_bin10_TTTo2L2Nu
449 prop_binCROS_electron2016M_bin7_Fake
450 prop_binSR_electron2016M_bin2_ZZtoLep
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
451 prop_binCROS_electron2017_bin7_WZ
452 prop_binCROS_electron2018_bin12_TTTo2L2Nu
453 prop_binCRF_electron2017_bin9_WpWpJJ_QCD
454 prop_binCRF_muon2017_bin8_ZZtoLep
455 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
456 prop_binSR_electron2016M_bin13_ZZtoLep
457 prop_binCRF_electron2017_bin13
458 prop_binCROS_electron2018_bin5_VG
459 prop_binSR_electron2016M_bin12_ZZtoLep
460 lep_2018
461 prop_binCRF_electron2017_bin9_Triboson
462 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
463 prop_binSR_electron2016M_bin12_WrongSign
464 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
465 prop_binSR_electron2016M_bin13_WrongSign
466 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
467 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
468 prop_binCRF_electron2016M_bin9_WpWpJJ_EWK
469 prop_binSR_electron2016M_bin2_TVX
470 prop_binCRF_electron2016M_bin7_Triboson
471 prop_binSR_electron2018_bin12_WrongSign
472 prop_binCROS_muon2018_bin13
473 prop_binSR_electron2016M_bin2_Triboson
474 prop_binCRF_electron2016M_bin7_ZZtoLep
475 prop_binCRF_electron2016M_bin9_Triboson
476 prop_binCRF_electron2016M_bin9_WpWpJJ_QCD
477 prop_binCRF_electron2016M_bin9_TVX
478 prop_binCROS_muon2017_bin8_TVX
479 prop_binCROS_electron2016M_bin13_Triboson
480 prop_binSR_muon2017_bin7_ZZtoLep
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.5 −0.4
481 prop_binCROS_electron2018_bin10_Triboson
482 prop_binCROS_muon2017_bin7_TVX
483 FES
484 prop_binCROS_electron2018_bin10_TVX
485 prop_binCROS_electron2016M_bin12_WpWpJJ_EWK
486 prop_binCROS_muon2017_bin7_VG
487 prop_binCROS_electron2016M_bin12_Triboson
488 prop_binCROS_electron2016M_bin7_Triboson
489 prop_binCROS_electron2018_bin12_Triboson
490 prop_binCROS_electron2018_bin5_TVX
491 prop_binCROS_muon2017_bin7_Triboson
492 prop_binCROS_muon2017_bin8_Triboson
493 prop_binCROS_electron2017_bin7_VG
494 prop_binCRF_electron2016M_bin9_ZZtoLep
495 prop_binCROS_electron2016M_bin13_ZZtoLep
496 prop_binCROS_electron2017_bin7_Triboson
497 prop_binCROS_electron2016M_bin13_WpWpJJ_EWK
498 prop_binCROS_electron2016M_bin12_TVX
499 prop_binCROS_electron2018_bin10_WpWpJJ_EWK
500 prop_binCROS_electron2018_bin10_ZZtoLep
501 prop_binCROS_electron2018_bin5_Triboson
502 prop_binCROS_electron2016M_bin12_ZZtoLep
503 prop_binCROS_electron2017_bin7_TVX
504 prop_binCROS_electron2016M_bin13_TVX
505 prop_binCROS_electron2016M_bin7_TVX
506 prop_binCROS_electron2017_bin7_ZZtoLep
507 prop_binCROS_electron2018_bin5_ZZtoLep
508 prop_binCROS_electron2018_bin12_TVX
509 QCDScale_WZ
510 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
511 prop_binCROS_electron2016M_bin7_ZZtoLep
512 prop_binCROS_electron2017_bin7_WpWpJJ_EWK
513 prop_binCROS_electron2018_bin12_WpWpJJ_EWK
514 prop_binCROS_electron2018_bin12_ZZtoLep
515 prop_binCROS_electron2018_bin5_WpWpJJ_EWK
516 prop_binCROS_muon2017_bin7_WpWpJJ_EWK
517 prop_binCROS_muon2017_bin7_ZZtoLep
518 prop_binCROS_muon2017_bin8_ZZtoLep
519 tau_vsele_2016M
520 tau_vsele_2017
521 tau_vsele_2018
522 tau_vsmu_2016M
523 tau_vsmu_2017
524 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$